

Hierarchical Risk Parity, Matrix Denoising & Industry Momentum

Comprehensive Quantitative Research & Portfolio Optimization Analysis

Note: This is an AI-generated summary of an ongoing research initiative by Daniel Paxton, which has been thoroughly reviewed.

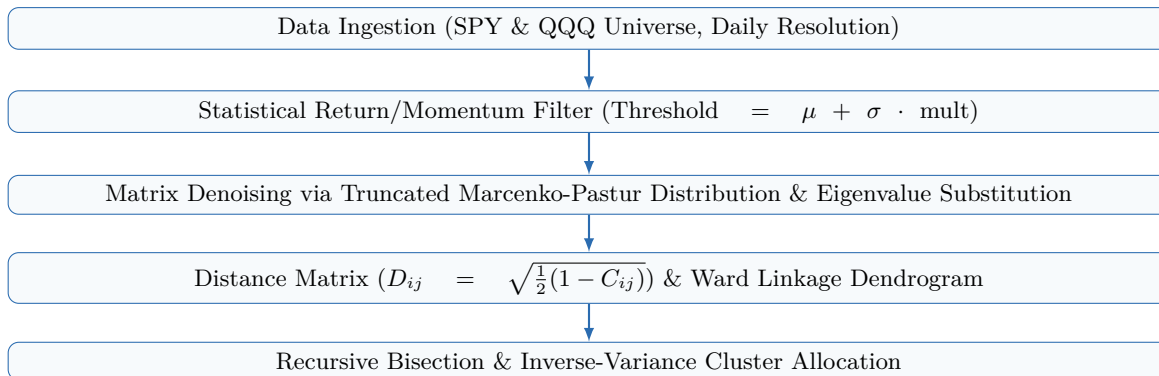
Page 1: Theoretical Methodology & Filter Architecture

1. Technical Overview & Academic Foundations

This report outlines an advanced quantitative portfolio construction framework implementing **Hierarchical Risk Parity (HRP)**, **Random Matrix Theory (RMT) Denoising**, and a **Statistical Cross-Sectional Return Filter**. The mathematical and statistical foundations integrate key insights from:

- Marcos López de Prado** – *Machine Learning for Asset Managers* (2020): Theoretical framework for Marcenko-Pastur covariance denoising, distance matrix mapping, and hierarchical cluster tree allocation.
- Tobias J. Moskowitz & Mark Grinblatt** – *Do Industries Explain Momentum?* (*The Journal of Finance*, DOI: 10.1111/0022-1082.00146): Establishes that financial momentum is heavily driven by industry and sector factors rather than idiosyncratic stock behavior, justifying industry-level pre-filtering and structural sector clustering.

2. Portfolio Pipeline Architecture



3. Deep Dive: The Statistical Asset & Momentum Filter

- Filter Mechanics:** Before performing covariance matrix estimation, the strategy applies a strict cross-sectional threshold:

$$\text{Threshold} = \mu_{\text{latest}} + (\sigma_{\text{latest}} \cdot \text{std_mult})$$

where μ_{latest} and σ_{latest} represent the cross-sectional mean and standard deviation of daily asset returns across the universe in the most recent estimation window. Assets falling below this threshold are pruned.

- Rationale & Industry Momentum Linkage:** As demonstrated by Moskowitz & Grinblatt (1999/2002), momentum strategies profit primarily due to persistent industry factors. Pruning assets below the threshold isolates equities exhibiting positive industry-driven momentum while filtering out persistent underperformers, effectively removing most low momentum assets per window.

1. Truncated Matrix Denoising Framework

Empirical covariance matrices derived from financial time series contain significant noise due to finite sample sizes (T) relative to asset count (N). Following López de Prado:

1. The empirical correlation matrix C is decomposed into eigenvalues λ_k and eigenvectors v_k .
2. The theoretical noise limit λ_+ is derived from the Marcenko-Pastur distribution:

$$\lambda_{\pm} = \sigma^2 \left(1 \pm \sqrt{\frac{N}{T}} \right)^2$$

where noise variance σ^2 is solved via Kernel Density Estimation (KDE) and non-linear optimization.

3. **Truncated Eigenvalue Processing (Three-Matrix Box Diagnostic):** Eigenvalues are strictly partitioned into two sets: **signal eigenvalues** ($\lambda_k > \lambda_+$) and **noise eigenvalues** ($\lambda_k \leq \lambda_+$). The noise eigenvalues are **truncated and replaced by their mean value** $\bar{\lambda}_{\text{noise}}$:

$$\tilde{\Lambda}_k = \begin{cases} \lambda_k & \text{if } \lambda_k > \lambda_+ \\ \bar{\lambda}_{\text{noise}} & \text{if } \lambda_k \leq \lambda_+ \end{cases}$$

The denoised correlation matrix is reconstructed via $\tilde{C} = V\tilde{\Lambda}V^T$ and re-scaled to ensure unit diagonals.

2. Visual Diagnostics: Three-Box Matrix Framework

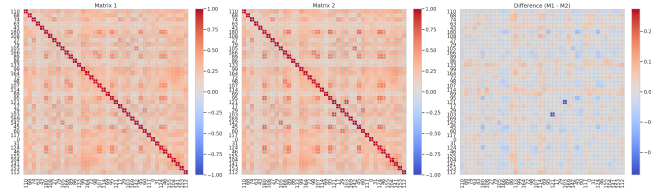
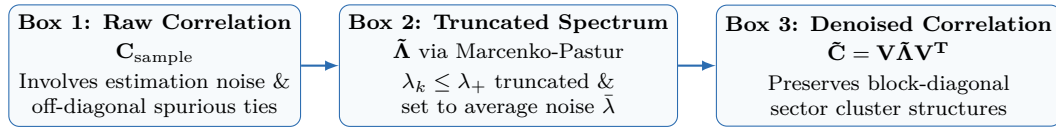


Figure 1: Three Box Matrix Framework Visual

3. Out-of-Sample Walk-Forward Results (2010–2026)

Performance Metric	Denoised HRP Walk-Forward	SPY Benchmark (2010–2026)
Annual Return (CAGR)	9.23%	14.08%
Cumulative Return	252.51%	783.66%
Annualized Volatility	13.99%	17.12%
Sharpe Ratio	0.70	0.86
Sortino Ratio	0.99	1.21
Maximum Drawdown	-29.86%	-33.72%
Equity Curve Stability	0.965	0.99